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MILITARY SPECIFICATION

COMPUTER TEST SET

(For Torpedo Mk 46)

This Specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers requirements for the procurement of one type of computer test set (see 6.1.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

Naval Ordnance Systems Command

DL 2540430	Computer Test Set, and all documents listed thereon

PUBLICATIONS

Naval Ordnance Systems Command

WD 28934	Special Manufacturing Procedure for Calibration of Computer Test Set, DL 2540430

FSC 1355

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(Copies of drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Basic requirements

3.1.1 Contractor-furnished data. No data is required by this specification or by applicable documents referenced in Section 2, unless specified in the contract or order (see 6.2).

3.1.2 Materials and treatments of materials and parts. Unless otherwise specified in the contract or order, all materials and treatments of materials and parts used in manufacture of the computer test set shall be in accordance with requirements specified herein and with the applicable documents listed in NAVORD DL 2540430.

3.2 Performance and product characteristics

3.2.1 Input power. The input power requirements of the computer test set (herein called test set) shall be as specified in Table I.

Table I

AC Input Power

Voltage	Current
115 ±10 volts alternating current (vac)	5 amperes maximum (amps max)

3.2.2. Test set outputs. The -40 volt power supply output regulation shall be such that a 40-ohm load resistance applied to the output shall not vary the voltage by more than 0.25 volt. The +25 volt power supply output regulation shall be such that a 150 ohm load resistance applied to the output shall not vary the voltage by more than 0.25 volt. The +25 volt power supply output shall furnish the +15 and +5 vdc sources and the -40 volt power supply output shall furnish the -15 vdc source to the computer under test.

3.2.3 Performance requirements. The test shall provide the following functions:

3.2.3.1 Test voltage. The test set test output voltages under specified load conditions shall be -15.0 ±0.1 vdc, +15.0 ±0.1 vdc, +5.0 ±0.15 vdc.

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3.2.3.2 Switching facilities

(a) The test set shall have panel switches for switching the following operations on and off: ODN TRIGGER, SNAKE SEARCH, COMPUTER ENABLE, PITCH RANGE CONSTRAINT, a VALID ECHO WIDTH switch with NORM and MIN positions; a YAW LEVEL switch with OFF plus four ON positions; a PITCH LEVEL switch with OFF and four ON positions; a TARGET POLARITY switch with UL and DR positions; a LAUNCH PROGRAM switch with AIR/ASROC and TUBE positions; a FIRE CONTROL switch ACTIVE and PASSIVE positions; a DOPPLER switch with -K OFF and +K positions; a TARGET switch with OFF, ON, and AT positions; a LEVEL CALIB pushbutton switch for PITCH and YAW DATA outputs.

3.2.3.3 Adjustment facilities. The test set front panel shall have adjustments for RANGE adjust, plus 15v adjust and minus 15v adjust; Y1, Y2, Y3; P1, P2, P3.

3.2.3.4 Test jack functions. The following test jacks with indicator lamps shall provide signals and display conditions listed in Table II during a test operation as follows:

Table II

Test Jack and Panel Lamp Requirements

Test Panel Nomenclature	Lamp Indication	Test Jack Signal
V E INHIBIT	Lamp illuminates	-12.0 vdc external input
VALID ECHO	Lamp flashes	See 3.2.3.5 (VALID ECHO WIDTH SWITCH to NORM)
PASSIVE MODE	Lamp illuminates	-12.0 vdc external input
SAMPLE	Lamp illuminates	-12.0 vdc external input
PING DRIVE	Lamp extinguishes	at ground level
RANGE GATE	Lamp illuminates	-12.0 vdc external input
RECEIVER PURSUIT HOLD	Lamp extinguishes	at ground level
+15v	No lamp	+15v power supply output
-15v	No lamp	-15v power supply output
REINSERT PENDULUM	Lamp extinguishes	-12.0 vdc external input

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Table II (Cont'd)

Test Jack and Panel Lamp Requirements

Test Panel Nomenclature	Lamp Indication	Test Jack Signal
NOT REATTACK	Lamp illuminates	-12.0 vdc external input
R<C	Lamp illuminates	-12.0 vdc external input
BLANK	Lamp illuminates	-12.0 vdc external input
R>B	Lamp extinguishes	at ground level
PITCH ENABLE	Lamp illuminates	-12.0 vdc external input
COMPUTER ENABLE	Lamp illuminates	See 3.2.3.16 (switch ON)
SNAKE SEARCH	Lamp illuminates	See 3.2.3.13 (switch ON)
ODN INITIATE	Lamp extinguishes	at ground level
SHIFT BEAM AXIS	Lamp extinguishes	-12.0 vdc external input
BEAM WIDTH FUNCTION	Lamp illuminates	-12.0 vdc external input
END SNAKE	Lamp illuminates	-12.0 vdc external input
AMPLITUDE THRESH	Lamp illuminates	See 3.2.3.8
COM (3jacks)	No lamp	Remote Sensing Grounds, J3201J
DOPPLER OFF PASSIVE MODE -12 vdc input, DOPPLER switch on +K and -K	No lamp	-12.0 vdc min. pulse
YAW DATA	No lamp	See 3.2.3.9
PITCH DATA	No lamp	See 3.2.3.9
+YAW (-YAW grounded)	No lamp	58 K load
-YAW (+YAW grounded)	No lamp	58 K load

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Table II (Cont'd)

Test Jack and Panel Lamp Requirements

Test Panel Nomenclature	Lamp Indication	Test Jack Signal
+PITCH (-PITCH grounded)	No lamp	58 K load
-PITCH (+PITCH grounded)	No lamp	58 K load
ANDED INVERTER	No lamp	75 K load
NOT SAMPLE KNOCKOUT	No lamp	30 K load
R<B	No lamp	-15V from 3 K source
R>B WARSHOT	No lamp	Continuity to J2101-P
STEERING READOUT	No lamp	Continuity to J2101-d

3.2.3.5 Valid echo pulse. A valid echo pulse having a minimum amplitude of -12 volts, a duration of 75 ±10 milliseconds and a maximum rise and fall time of 5 microseconds shall be displayed when VALID ECHO WIDTH switch is positioned to NORM and TARGET switch is positioned to ON. A valid echo pulse shall not be present when TARGET switch is positioned to OFF or when valid echo inhibit signal is present. A valid echo pulse of 0.9 ±0.1 millisecond shall be displayed when VALID ECHO WIDTH switch is placed to MIN.

3.2.3.6 Valid echo range. When the RANGE dial is set at exactly 5.00 the time interval between the leading edge of PING DRIVE output and leading edge of VALID ECHO output shall be 607.5 ±7.5 milliseconds.

3.2.3.7 Valid echo inhibit. When -12 volts stimulus is applied to VALID ECHO INHIBIT input, the VALID ECHO output pulse shall cease. When stimulus voltage and TARGET are removed, VALID ECHO output shall be from +0.2 to -0.2 volt.

3.2.3.8 Amplitude threshold. An amplitude threshold output voltage of -12 volts minimum shall be present coincident with valid echo pulse (3.2.3.5) when TARGET switch is positioned to AT.

3.2.3.9 Yaw and pitch data. With the TARGET switch positioned to ON, the LEVEL CALIB pushbutton depressed, and YAW and PITCH LEVEL switches positioned to 1, 2, and 3, the following voltages shall be displayed on digital dc voltmeter for the indicated conditions.

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(a) For TARGET POLARITY switch at DR position:

YAW DATA voltages shall be:

-1.10 ±0.01 vdc at YAW LEVEL 1
-2.20 ±0.02 vdc at YAW LEVEL 2
-4.00 ±0.50 vdc at YAW LEVEL 3

PITCH DATA voltages shall be:

+1.10 ±0.01 vdc at PITCH LEVEL 1
+2.20 ±0.02 vdc at PITCH LEVEL 2
+4.00 ±0.50 vdc at PITCH LEVEL 3

(b) For TARGET POLARITY switch at UL position:

YAW DATA voltages shall be:

+1.10 ±0.01 vdc at YAW LEVEL 1
+2.20 ±0.02 vdc at YAW LEVEL 2
+4.00 ±0.50 vdc at YAW LEVEL 3

PITCH DATA voltages shall be:

-1.10 ±0.01 vdc at PITCH LEVEL 1
-2.20 ±0.02 vdc at PITCH LEVEL 2
-4.00 ±0.50 vdc at PITCH LEVEL 3

(c) With TARGET switch positioned to OFF, PITCH DATA and YAW DATA voltages shall not exceed 100 millivolts.

3.2.3.10 Passive mode. With -12 volts applied to PASSIVE MODE and TARGET switch positioned to ON or AT, the VALID ECHO output shall indicate a negative dc voltage. With no voltage applied to PASSIVE MODE, the VALID ECHO output shall indicate a pulsed voltage.

3.2.3.11 Doppler. With TARGET switch positioned to ON or AT and DOPPLER switch positioned to +K, the +K output voltage shall be -12 vdc min. With DOPPLER switch positioned to -K, the -K output voltage shall be -12 vdc min. With TARGET switch positioned to OFF or DOPPLER switch positioned to OFF the ±K voltages shall be 0.0 ±0.2 vdc.

3.2.3.12 Fire control. With FIRE CONTROL switch positioned to ACTIVE and LAUNCH PROGRAM switch positioned to TUBE, TUBE LAUNCH output voltage shall be -15.0 ±0.3 vdc. The same voltage indication applies for ACTIVE output.

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3.2.3.13 Snake search. With the LAUNCH PROGRAM switch in TUBE and with SNAKE SEARCH switch ON, snake search output shall be 0.0 ±0.2 vdc. With the SNAKE SEARCH switch OFF, the output shall be -15.0 ±0.3 vdc.

3.2.3.14 ODN trigger. With ODN TRIGGER switch to OFF, ODN TRIGGER output voltage shall be -15.0 ±0.3 vdc. With the ODN TRIGGER switch to ON, the ODN TRIGGER output voltage shall be 0.0 ±0.2 vdc.

3.2.3.15 Pitch range constraint test. With PITCH RANGE CONSTRAINT switch to ON, PITCH RANGE CONSTRAINT output voltage shall be -15.0 ±0.3 vdc. With the PITCH RANGE CONSTRAINT switch OFF, the PITCH RANGE CONSTRAINT line shall be open circuited.

3.2.3.16 Computer enable. When COMPUTER ENABLE switch in positioned from OFF to ON, an output voltage of -15 vdc shall be obtained. With the COMPUTER ENBALE switch OFF, the output shall be 0.0 ±0.2 vdc.

3.3 Workmanship. Workmanship and finish shall be of sufficiently high grade to assure satisfactory operation consistent with requirements of this specification. All parts used in the test set shall be thoroughly clean and free of excess materials, chips, and loose or spattered foreign materials. Standards of workmanship, subject to any qualifications stated in the Government's notice of approval, shall be determinative of requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections of the test set shall be classified as follows:

4.2.1 Quality conformance inspection

(a) Quality conformance sampling (see 4.3.1)

(b) Quality conformance tests and Examinations (see 4.5)

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4.3 Sampling

4.3.1 Quality conformance sampling. The tests and examinations specified in 4.5 shall be conducted on each test set to be delivered; therefore, there shall be no sampling plan for acceptance. Defective units shall be rejected.

4.4 Test conditions. Unless otherwise specified, the test set shall be subjected to tests under the following conditions:

(a)	Temperature	Room ambient (60 to 95 °Fahrenheit (F)
(b)	Humidity (Relative)	Room ambient, up to 95 percent
(c)	Altitude	Normal ground atmospheric pressure

4.5 Quality conformance tests and examinations. Unless otherwise specified, each test set delivered under contract shall be subjected to the tests and examinations contained in 4.5.3.

4.5.1 Special notes

4.5.1.1 Calibration. Calibration of the test set should be performed in accordance with WD 28934 and before conducting tests in 4.5.3.

4.5.1.2 Test sequence. Unless otherwise noted, sequence of testing precedures presented herein is mandatory and must be performed in the order shown.

4.5.1.3 Test results. Where space is provided in Appendix Report Form to indicate value or specific reading obtained, the specific reading shall be recorded if a limit or limits are given in Limits column for the test. If limits are not stated in Limits column, the individual performing the test or inspection shall initial in space provided in the Reading column indicating that requirements were met satisfactorily.

4.5.1.4 Acceptance criteria. Failure of the end product to pass any examination or test of this section will automatically classify the unit as nonconforming. When nonconforming units are corrected by the contractor, the unit shall be reinspected. The reinspection of a nonconforming unit may be limited to the test or examination which defined the nonconformance, or when so directed by the cognizant inspector, a complete reexamination and retest of the unit may be required. Nonconforming units, which have not been corrected, will be considered for acceptance only upon formal application by the contractor to the Naval Ordnance Systems Command via the cognizant Government representative.

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4.5.2 Quality conformance test equipment. Items of test equipment shown in Table III are required to perform quality conformance tests specified herein. All items of test equipment shall be of sufficient accuracy and precision to permit valid performance of required tests. Test instruments shall be calibrated at regular intervals and procedures for and results of all calibrations shall be subject to approval of the cognizant Government Inspector. Items having equivalent characteristics may be substituted where necessary.

Table III

Test Equipment

Item	Identification
Power Supply: (Regulated output; 0 to 30 vdc, 0 to 150 ma dc; Load Regulation less than 0.3% from 0 to 150 ma; Line regulation less than 0.3% for ±10% line voltage change; Noise and Ripple less than 150 microvolts)	Hewlett-Packard model 721A
Oscilloscope; (Bandpass from dc to 15 megacycles; Rise time 0.023 microseconds; Accuracy 3% from 0.05 to 20 volts/cm)	Tektronix model 535A w/type CA preamplifier
Electronic Counter: (Range from 10 Hertz (Hz) to 120 kiloHertz (kHz) Accuracy to ±1 count)	Hewlett-Packard model 522B
Digital dc Voltmeter; (Range: 0.001 to 999 vdc; Accuracy: ±0.2% ±1 count. Input impedance 11 megohms.)	Hewlett-Packard Model 405BR
Multimeter: (Resistance from 0 to 20 megohms; Current from 0 to 10 amperes; Voltage 0 to 5000 dc or ac; Accuracy ±3% full scale for dc, ±5% full scale for ac)	Simpson model 260
Resistor: (20 ohms, 25 watts ±5%)	

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4.5.3 Test procedure

4.5.3.1 -40 Volt power supply setup

4.5.3.1.1 Position POWER switch, and D. C. POWER switch located on power panel, to ON. Adjust power supply to obtain -39.8 to -40.2 vdc as indicated on digital DC voltmeter. Record meter indication.

4.5.3.2 +25 Volt power supply setup

4.5.3.2.1 Adjust VOLTAGE control located on +25 volt power supply panel, to obtain +24.8 to +25.2 vdc as indicated on digital DC voltmeter. Record meter indication.

4.5.3.3 Power supply test

NOTE: Test points designated by a cable connector number followed by a pin number are located on two cable connectors (J3201 and J2101) at end of test cable plugged into test set connector panel.

4.5.3.3.1 Connect jumper from J2101-X to J3201-J. Connect jumper from J2101-c to J3201-M. Jumpers shall remain in place for remainder of test.

4.5.3.3.2 Using an ohmmeter, check resistance between pins T and U on J2101. Indication shall be 1 ohm max.

4.5.3.3.3 Using digital dc voltmeter, make following measurements and adjustments. Use J2101-X (COMMON) as ground return.

4.5.3.3.4 Monitor voltage at J3201-R (+15 vdc) and adjust +15V ADJUST control on test set test panel for a voltage indication of +15.0 ±0.1 vdc. Monitor voltage at J2101-W (+5 vdc). Voltage indication shall be 5.00 ±0.15 vdc. Record meter indications.

4.5.3.3.5 Connect a 20 ohm 25 watt resistor between J2101-V and J2101-X. Monitor voltage at J2101-V (-15 vdc) and adjust -15V ADJUST control on test set test panel to obtain a voltmeter indication of -15.0 ±0.1 vdc. Resistor shall remain in place for remainder of test.

4.5.3.4 Panel indicator lamps test

4.5.3.4.1 Connect positive (+) terminal of the external power supply to COM jack on test set panel. Position TARGET switch on panel of OFF.

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4.5.3.4.2 When -12.0 volts from the external power supply is connected to the pins listed below the panel lamp listed on the same line shall illuminate. Record lamp condition (illuminated or extinguished).

NOTE: Some panel lamps and test jacks have two function names because of the test set capability of testing both computers for Mk 46 Mods 0 and 1, and computer for Mk 46 Mod 2. When referring to these lamps and jacks the MOD 0/1 terminology, which is marked in white on the panel, will be used.

Pin	Panel Lamp
J2101-B	BLANK
J2101-C	SAMPLE
J2101-E	RANGE GATE
J2101-F	NOT REATTACK
J2101-J	V E INHIBIT
J2101-L	PASSIVE MODE
J2101-M	BEAM WIDTH FUNCTION
J2101-S	R<C
J3201-N	PITCH ENABLE
J3201-BB	END SNAKE

4.5.3.4.3 The following illuminated panel lamps shall extinguish when -12.0 volts is applied to the indicated pin. Record lamp condition (illuminated or extinguished).

Pin	Panel Lamp
J2101-N	SHIFT BEAM AXIS
J3201-X	REINSERT PENDULUM

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4.5.3.4.4 The following illuminated panel lamps shall extinguish when the indicated pin is grounded (J2101-X). Record lamp condition (illuminated or extinguished).

Pin	Panel Lamp
J2101-A	PING DRIVE
J2101-D	RECEIVER PURSUIT HOLD
J2101-R	ODN INITIATE
J2101-b	R > B

4.5.3.5 Valid echo pulse test

4.5.3.5.1 Connect a jumper from panel COM jack to J2101-J (VALID ECHO INHIBIT). Remove PING DRIVE indicator lamp from panel socket. Adjust RANGE dial on panel to 5.00. Position TARGET switch to ON. Connect oscilloscope input lead to J3201-C (VALID ECHO). Position VALID ECHO WIDTH switch to MIN. VALID ECHO indicator lamp shall not flash intermittently. Record lamp condition (flashing or not flashing). Pulse displayed by oscilloscope shall be 0.9 ±0.1 millisecond. Record pulse.

4.5.3.5.2 Position VALID ECHO WIDTH switch to NORM. VALID ECHO indicator lamp shall flash on and off continuously. Record lamp condition (flashing or not flashing). Pulse displayed on oscilloscope shall have a minimum amplitude of -12 volts, 75 ±10 milliseconds duration and 5 microseconds maximum rise and fall time. Record pulse amplitude, duration, and rise and fall time.

4.5.3.6 Valid echo range test

4.5.3.6.1 Position TRIGGER INPUT on electronic counter to SEP position. Connect J2101-A (PING DRIVE) to START input and J3201-C (VALID ECHO) to STOP input of counter. Position both TRIGGER SLOPES on counter to negative (-) and TRIGGER LEVEL VOLTS for each counter input to -3. Ground PING DRIVE jack on test set beiefly and observe counter. Interval displayed shall be 607.5 ±7.5 milliseconds with test set RANGE dial set at exactly 5.00. Record indicated time interval.

4.5.3.7 Amplitude threshold test

4.5.3.7.1 Connect oscilloscope input lead to J3201-D (AMPLITUDE THRESHOLD) and verify that target pulse is not present on this pin except when test set TARGET switch is set to AT. With TARGET switch in AT position, AMPLITUDE target pulse is present also at J3201-C (VALID ECHO). Both VALID ECHO and AMPLITUDE THRESH indicator lamps shall flash on and off. Record indications. With TARGET switch in ON or OFF, voltage at J3201-D shall be

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0.0 ±0.2 vdc. Remove jumper between J2101-J and panel COM jack.

4.5.3.8 Valid echo inhibit test

4.5.3.8.1 Position TARGET switch to AT. Apply -12 volts to J2101-J (VALID ECHO INHIBIT). Pulse at J3201-C (VALID ECHO) shall cease. Remove -12 volt stimulus and position TARGET switch to OFF. J3201-C (VALID ECHO) shall measure from +0.2 to -0.2 volt. Record indications.

4.5.3.9 Yaw data test

4.5.3.9.1 Position TARGET switch to ON and MOD switch to 0/1.

4.5.3.9.2 Connect digital dc voltmeter to YAW DATA test jack.

4.5.3.9.3 Position TARGET POLARITY, and YAW LEVEL switches as indicated in Table IV. Depress and hold LEVEL CALIB switch for indication. Readjust Y1, Y2, or Y3 if necessary. Record indication.

Table IV

Yaw Data Test

YAW LEVEL Switch	Adjust	TARGET POLARITY Switch Position	Voltage Indication
1	Y1	DR	-1.10 ±0.01 vdc
2	Y2	DR	-2.20 ±0.02 vdc
3	Y3	DR	-4.00 ±0.50 vdc
1	Y1	UL	+1.10 ±0.01 vdc
2	Y2	UL	+2.20 ±0.02 vdc
3	Y3	UL	+4.00 ±0.50 vdc

4.5.3.10 Pitch data test

4.5.3.10.1 Connect digital dc voltmeter to PITCH DATA test jack.

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4.5.3.10.2 Position TARGET POLARITY and PITCH LEVEL switches as indicated in Table V. Depress and hold LEVEL CALIB switch for indication. Readjust P1, P2, or P3 if necessary. Record indication.

Table V

Pitch Data Test

PITCH LEVEL Switch Position	Adjust	TARGET POLARITY Switch Position	Voltage Indication
1	P1	UL	-1.10 ±0.01 vdc
2	P2	UL	-2.20 ±0.02 vdc
3	P3	UL	-4.00 ±0.50 vdc
1	P1	DR	+1.10 ±0.01 vdc
2	P2	DR	+2.20 ±0.02 vdc
3	P3	DR	+4.00 ±0.50 vdc

4.5.3.11 Polarity tests

4.5.3.11.1 Connect oscilloscope to J3201-B (PITCH DATA) and synchronize from VALID ECHO jack. Oscilloscope shall display a positive pulse greater than 3.5 volts in amplitude. The pulse shall reach 2/3 (67%) of its maximum amplitude in 2.5 to 4.2 msec. Record pulse amplitude and time constant.

4.5.3.11.2 Position TARGET POLARITY to UL. Oscilloscope shall display a negative pulse greater than 3.5 volts in amplitude. The pulse shall reach 2/3 (67%) of its maximum amplitude in 2.5 to 4.2 msec. Record pulse amplitude and time constant.

4.5.3.11.3 Connect oscilloscope to J3201-A (YAW DATA). Oscilloscope shall display a positive pulse greater than 3.5 volts in amplitude. The pulse shall reach 2/3 (67%) of its maximum amplitude in 2.5 to 4.2 msec. Record pulse amplitude and time constant.

4.5.3.11.4 Position TARGET POLARITY to DR. Oscilloscope shall display a negative pulse greater than 3.5 volts in amplitude. The pulse shall reach 2/3 (67%) of its maximum amplitude in 2.5 to 4.2 msec. Record pulse amplitude and time constant.

4.5.3.12.1 Apply -12 volts to PASSIVE MODE jack on test panel. VALID ECHO and PASSIVE MODE indicator lamps shall illuminate. Voltage at

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J3201-C (VALID ECHO) and J3201-E (-K) shall be -12 vdc minimum with DOPPLER switch set at -K. Voltage at J2101-H (+K) shall be -12 vdc minimum with DOPPLER switch at +K. Record indications.

4.5.3.12.2 Position the TARGET switch to OFF. Voltage measured at J3201-B (PITCH DATA) and J3201-A (YAW DATA), shall not exceed 100 millivolts for TARGET POLARITY switch in UL or DR position. J3201-C (VALID ECHO), J3201-E (-K) and J2101-H (+K) shall not exceed 200 millivolts. Record indication.

4.5.3.12.2 Remove -12 volts dc stimulus from PASSIVE MODE jack. Position TARGET switch to ON. Connect oscilloscope input lead to J2101-H (+K). Determine that a -12 volt minimum pulse is present, coincident with VALID ECHO pulse. Set DOPPLER switch to -K. Determine -12 volt minimum pulse is present coincident with VALID ECHO pulse at J3201-E. Record indications.

4.5.3.13 Fire control test

4.5.3.13.1 Position FIRE CONTROL switch to ACTIVE position and LAUNCH PROGRAM switch to TUBE position. Digital dc voltmeter indication at J3201-Y (TUBE LAUNCH PROGRAM) shall be -15.0 ±0.3 vdc. Record indication.

4.5.3.13.2 Position SNAKE SEARCH switch to OFF. Voltage at J3201-Z shall be -15.0 ±0.3 vdc.

4.5.3.13.3 Position SNAKE SEARCH switch to ON. SNAKE SEARCH indicator lamp shall illuminate. Record lamp condition (illuminated or extinguished). Voltage at J3201-Z shall be +0.2 to -0.2 vdc.

4.5.3.13.4 Position LAUNCH PROGRAM switch to AIR/ASROC. Voltage at J3201-Y shall be +4.5 to +5.5 vdc. Set SNAKE SEARCH switch to OFF. After 30 seconds, voltage at J3201-Y shall be +1.0 to -1.0 vdc.

4.5.3.13.5 Measure voltage at J2101-k. Voltage indication shall be -15.0 ±0.3 vdc. Record indication.

4.5.3.13.6 Measure voltage at J3201-AA. After 30 seconds, voltage indication shall be +1.0 to -1.0 vdc. Record indication.

4.5.3.13.7 Position FIRE CONTROL switch to PASSIVE. Measure voltage at J3201-AA. Voltage indication shall be -15.0 ±0.3 vdc. Record indication.

4.5.3.13.8 Measure voltage at J2101-k. After 30 seconds, voltage indication shall be +1.0 to -1.0 vdc. Record indication.

4.5.3.13.9 Measure voltage at J3201-CC. Voltage indication shall be -15.0 ±0.3 vdc.

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4.5.3.14 ODN trigger test

4.5.3.14.1 Position ODN TRIGGER switch to OFF. Connect digital dc voltmeter to J2101-n. Voltage indication shall be -15.0 ±0.3 vdc. Record indication.

4.5.3.14.2 Position ODN TRIGGER switch to ON. Voltage indication shall be +0.2 to -0.2 vdc. Record indication.

4.5.3.15 Pitch range constraint test

4.5.3.15.1 Position PITCH RANGE CONSTRAINT switch to ON. Connect digital dc voltmeter to J3201-P. Voltage indication shall be -15.0 ±0.3 vdc. Record indication.

4.5.3.15.2 Position PITCH RANGE CONSTRAINT switch to OFF. After 30 seconds, voltage indication shall be +1.0 to -1.0 vdc. Record indication.

4.5.3.16 Computer enable test

4.5.3.16.1 Connect oscilloscope input lead to J3201-W (COMPUTER ENABLE). When COMPUTER ENABLE switch is turned from OFF to ON, voltage observed on oscilloscope screen shall step from 0 volts to -15 volts. Record indication.

4.5.3.16.2 Position COMPUTER ENABLE switch to ON position. COMPUTER ENABLE indicator lamp shall illuminate. Record lamp condition (illuminated or extinguished). Position COMPUTER ENABLE switch to OFF. Voltage at J3201-W shall be +0.2 to -0.2 vdc. Record indication.

4.5.3.17 Test jack continuity

4.5.3.17.1 Set DC POWER switch and POWER switch located on power panel to OFF. Connect multimeter (RX10K scale) between the following test jack and a GRD jack. Indications shall be as specified in Table VI. Record indication.

Table VI

Test Panel Jack	Indication
+YAW	56 K to 62 K
-YAW	56 K to 62 K
+PITCH	56 K to 62 K
-PITCH	56 K to 62 K
ANDED INVERTER	70 K to 80 K
NOT SAMPLE KNOCKOUT	27 K to 33 K

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4.5.3.17.2 Check continuity between the following test panel jacks and connector pins. The indications shall be as specified. Record results.

Test Panel Jack	Connector Pin	Indication
R > B WARSHOT	J-2101-P	1 ohm max.
STEERING READOUT	J2101-d	1 ohm max.

4.5.3.17.3 Test jack continuity. Measure resistance bwtween R < B test jack on test panel and -15V jack on test panel, using multimeter (RX100 scale). Indication shall be 2.7 K to 3.3 K. Record indication.

4.5.3.18 MOD 2 function tests

4.5.3.18.1 Set MOD switch to 2. With multimeter, measure resistance to COM (J2101-X) at each of the following front panel jacks: +YAW and + PITCH. In each case resistance shall be between 56,000 and 62,000 ohms. Record indication.

4.5.3.18.2 With multimeter, measure resistance between the -PITCH and the -15 VDC front panel jacks. The resistance shall be between 2150 and 2250 ohms. Record indication.

4.5.3.18.3 Set SD NOT ACHIEVED switch to OFF. With multimeter measure resistance between -YAW front panel jack and COM (J2101-X). The resistance shall be 1 ohm maximum. Record indication.

4.5.3.18.4 Set SD NOT ACHIEVED switch to ON. With multimeter, measure resistance between the -YAW and the V E INHIBIT front panel jacks. Resistance shall be 1 ohm maximum. Record indication.

4.5.3.18.5 With multimeter, measure resistance between J3201-R (+15 VDC) and J3201-CC (NOT 6°). Resistance shall be 35,000 to 37,000 ohms. Record indication.

4.5.3.18.6 With multimeter, measure resistance between J2101-C (-15 VDC) and J3201-AA (NOT 10°). Resistance shall be 2640 to 2760 ohms. Record indication.

4.5.3.19 Test jack continuity. Set PITCH LEVEL and YAW LEVEL switches to OFF. Remove jumper between J2101-X and J3201-J. Connect multimeter to theCOM jack next to the ANDED INVERTER jack. Measure continuity to the 2 other COM jacks. Indication shall be 1 ohm max. Measure resistance between J2101-X and J3201-J. Resistance shall be greater than 4 K ohms. Record indications.

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4.5.3.20 Cable continuity. Measure the resistance between the following computer panel test jacks and the indicated connector pins. The resistance shall be 1 ohm max. Record indication.

Panel Test Jack	Connector Pins
+15VDC	J3201-R
-15VDC	J2101-c
COM	J3201-J
V E INHIBIT	J2101-J
VALID ECHO	J3201-C
PASSIVE MODE	J2101-L
BLANK	J2101-B
SAMPLE	J2101-C
PING DRIVE	J2101-A
RANGE GATE	J2101-E
RECEIVER PURSUIT HOLD	J2101-D
REINSERT PENDULUM	J3201-X
NOT REATTACK	J2101-F
R < C	J2101-S
R > B	J2101-b
PITCH ENABLE	J3201-N
COMPUTER ENABLE	J3201-W
SNAKE SEARCH	J3201-Z
ODN INITIATE	J2101-R
SHIFT BEAM AXIS	J2101-N
BEAM WIDTH FUNCTION	J2101-M

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4.5.3.20 (Cont'd)

Panel Test Jack	Connector Pins
NOT SAMPLE KNOCKOUT	J2101-K
R < B	J3201-H
END SNAKE	J3201-BB
AMPLITUDE THRESHOLD	J3201-D
YAW DATA	J3201-A
PITCH DATA	J3201-B
+YAW	J3201-S
-YAW	J3201-T
+PITCH	J3201-U
-PITCH	J3201-V
ANDED INVERTER	J3201-L

4.5.3.21 Remove all test leads, remove jumpers between J2101-c and J3201-M, remove 20Ω resistor and replace PING DRIVE indicator lamp removed during target function tests (4.5.3.5.1).

4.5.4 Visual inspections

4.5.4.1 Identification marking

4.5.4.1.1 The nomenclature, control callout, and identification data shall be correct, clear, and legible. (Reference: NAVORD DL 2540430.)

4.5.4.2 Interchangeability

4.5.4.2.1 The connector receptacles shall be correct and capable of accepting the mating units without resort to selection, modification, or fitting. (Reference: NAVORD DL 2540430.)

4.5.5 Process verification

4.5.5.1 Criteria stipulated in this paragraph are applicable to the production process instead of specific units of product. Inspections stipulated are general in scope and technique and frequency of verification will be

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contingent upon specific process conditions. Positive action by the contractor is required when process criteria findings indicate poor inspection by the contractor.

4.5.5.2 Lacing

4.5.5.2.1 Lacing and cable configuration shall be in accordance with procedures approved by the procuring activity. Specific methods of verifying the adequacy of cable lacing shall include visual and dimensional verification.

4.5.5.3 Soldering

4.5.5.3.1 Soldered joints shall be in accordnace with good commerical practices. Unsoldered connections or those of a "cold" solder or "rosin" joint shall be considered defective.

4.5.5.4 Cleanliness

4.5.5.4.1 Presence of foreign material in assembly shall classify unit as defective. Foreign material includes anything not intended to be present.

4.5.6 Packaging and packing inspections

4.5.6.1 Packaging, packing, and marking shall comply with Section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level C.

5.1.1 Level A. Not applicable.

5.1.2 Level B. Not applicable.

5.1.3 Level C. Cleaning, drying, preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity and may conform to the supplier's commercial practice.

5.2 Packing. Packing shall be level C as specified.

5.2.1 Level A. Not applicable.

5.2.2 Level B. Not applicable.

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5.2.3 Level C. The test set, packaged as specified in 5.1.3, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Marking. In addition to marking required by the contract or order, unit packages and shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use and description

6.1.1 Intended use. The test set is intended for use in a manufacturing facility during the quality conformance testing of computer for Torpedo Mk 46.

6.1.2 Description

6.1.2.1 Mechanical. The test set is a rectangular cabinet containing a rack mounted test panel and a rack mounted power supply.

6.1.2.2 Electrical. The test set simulates the electrical inputs and outputs of the computer.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Any exceptions to this specification.

(c) Required data.

(d) All additional samples required shall be provided and inspected at the expense of the supplier, when such additional samples are required because of failure of any sample(s) to pass the prescribed inspections.

6.3 Definitions. For the purpose of this specification, the following definitions have been established:

6.3.1 Operating. Operating is the condition of the test set performing its required function.

6.3.2 Temperature. The temperature of the environment to which the test set may be exposed, expressed in degrees Fahrenheit (°F).

6.3.3 Pressure. The pressure of the environment to which the test set may be exposed, expressed in pounds per square inch absolute (psia).

6.3.4 Humidity. The relative humidity of the environment to which the test set is exposed, expressed as a percent.

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REPORT FORM

QUALITY ASSURANCE PROVISION

QUALITY CONFORMANCE TESTS AND EXAMINATIONS

FOR

COMPUTER TEST SET

DL 2540430

(For Torpedo Mk 46)

EQUIPMENT STATUS			
CONTRACTOR CERTIFICATION		GOVERNMENT ACCEPTANCE	
Contractor		Date of Acceptance	
Contract		Accepted by	
Equipment Ser. No.		Date of Acceptance	
Insp. Req. Satisfied		Agency	

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REPORT FORM

QUALITY ASSURANCE PROVISION

QUALITY CONFORMANCE TEST AND EXAMINATION

COMPUTER TEST SET

10. SCOPE

10.1 This Report Form contains data sheets to record the results of the quality conformance tests and examinations of 4.5 of this specification. When completed with authorized certification and acceptance signatures affixed, this report form or copies thereof shall accompany the test set bearing the recorded equipment serial number.

20. GENERAL INSTRUCTIONS

20.1 Test sequence. The quality conformance tests and examinations of 4.5 shall be completed and the results recorded or verified in this report form prior to acceptance of the test set. Unless otherwise noted the sequence of testing, recording and verification, as performed in 4.5 is mandatory and must be performed in the order shown.

20.2 Recording of results. Where space is provided to indicate the value of specific reading obtained, the specific reading will be recorded if a limit or limits is given in the Limit column for the test. The initials of the individual performing the inspection shall be inserted above the value observed and recorded. If limits are not stated in the Limit column, the individual performing the test or inspection shall initial in the space provided in the Reading column indicating that the requirement has been met satisfactorily.

30. QUALITY CONFORMANCE TEST AND EXAMINATION

	Reference	Limits	Reading
30.1 Test equipment	4.5.2	In calibration	_____
30.2 Test procedure calibration per WD 28934	4.5.3	Completed	_____
30.2.1 -40 Volt power supply setup.	4.5.3.1		
-40 volt source	4.5.3.1.1	-39.8 to -40.2 vdc	_____
30.2.2 +25 Volt power supply setup.	4.5.3.2		
+25 volt source	4.5.3.2.1	+24.8 to +25.2 vdc	_____

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		Reference	Limits	Reading
30.2.3	Power supply test	4.5.3.3		
	Resistance, J2101-T to J2101-U	4.5.3.3.2	1 ohm max	_____
	+15 vdc indication	4.5.3.3.4	14.9 to 15.1 vdc	_____
	+5 vdc indication		4.85 to 5.15 vdc	_____
	-15 vdc indication	4.5.3.3.5	-14.9 to -15.1 vdc	_____
30.2.4 test	Panel indicator lamps	4.5.3.4		
	BLANK	4.5.3.4.2	Illuminated	_____
	SAMPLE		Illuminated	_____
	RANGE GATE		Illuminated	_____
	NOT REATTACK		Illuminated	_____
	V E INHIBIT		Illuminated	_____
	PASSIVE MODE		Illuminated	_____
	BEAM WIDTH FUNCTION		Illuminated	_____
	R < C		Illuminated	_____
	PITCH ENABLE		Illuminated	_____
	END SNAKE		Illuminated	_____
	SHIFT BEAM AXIS	4.5.3.4.3	Extinguished	_____
	REINSERT PENDULUM		Extinguished	_____
	PING DRIVE	4.5.3.4.4	Extinguished	_____
	RECEIVER PURSUIT HOLD		Extinguished	_____
	ODN INITIATE		Extinguished	_____
	R > B		Extinguished	_____

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		Reference	Limits	Reading
30.2.5	Valid echo pulse test	4.5.3.5		
	VALID ECHO	4.5.3.5.1	Not flashing	_____
	Oscilloscope pulse		0.8 to 1.0 ms	_____
	VALID ECHO	4.5.3.5.2	Flashing	_____
	Oscilloscope pulse amplitude		-12 volts min	_____
	Oscilloscope pulse duration		65 to 85 ms	_____
	Oscilloscope pulse rise and fall time		5 μs max	_____
30.2.6	Valid echo range test	4.5.3.6		
	Time interval	4.5.3.6.1	600 to 615 ms	_____
30.2.7	Amplitude threshold test	4.5.3.7		
	J3201-D (AMPLITUDE THRESH-HOLD)	4.5.3.7.1	Pulses only in AT	_____
	J3201-C (VALID ECHO)		Pulses in AT	_____
	VALID ECHO		Shall flash on and off	_____
	AMPLITUDE THRESHOLD		Shall flash on and off	_____
	J3201-D with TARGET ON or OFF		From +0.2 to -0.2 volts	_____
30.2.8	Valid echo inhibit test	4.5.3.8		
	VALID ECHO pulse	4.5.3.8.1	Shall cease	_____
	VALID ECHO output		From +0.2 to -0.2 volts	_____

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		Reference	Limits	Reading
30.2.9	Yaw data test	4.5.3.9		
	YAW DATA test jack	4.5.3.9.3		
	DR, YAW LEVEL Switch Position			
	1		-1.09 to -1.11 vdc	_____
	2		-2.18 to -2.22 vdc	_____
	3		-3.50 to -4.50 vdc	_____
	UL, YAW LEVEL Switch Position			
	1		+1.09 to +1.11 vdc	_____
	2		+2.18 to +2.22 vdc	_____
	3		+3.50 to +4.50 vdc	_____
30.2.10	Pitch data test	4.5.3.10		
	PITCH DATA test jack	4.5.3.10.2		
	UL, PITCH LEVEL Switch Position			
	1		-1.09 to -1.11 vdc	_____
	2		-2.18 to -2.22 vdc	_____
	3		-3.50 to -4.50 vdc	_____
	DR, PITCH LEVEL Switch Position			
	1		+1.09 to +1.11 vdc	_____
	2		+2.18 to +2.22 vdc	_____
	3		+3.50 to +4.50 vdc	_____

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		Reference	Limits	Reading
30.2.11	Polarity tests	4.5.3.11		
	Oscilloscope positive pulse amplitude	4.5.3.11.1	Positive greater than 3.5 volts	_____
	Pulse time constant		2.5 to 4.2 msec	_____
	Oscilloscope negative pulse amplitude	4.5.3.11.2	Negative greater than 3.5 volts	_____
	Pulse time constant		2.5 to 4.2 msec	_____
	Oscilloscope positive pulse amplitude	4.5.3.11.3	Positive greater than 3.5 volts	_____
	Pulse time constant		2.5 to 4.2 msec	_____
	Oscilloscope negative pulse amplitude	4.5.3.11.4	Negative greater than 3.5 volts	_____
	Pulse time constant		2.5 to 4.2 msec	_____
30.2.12	Passive mode test	4.5.3.12		
	VALID ECHO and PASSIVE MODE lamps	4.5.3.12.1	Illuminated	_____
	VALID ECHO voltage		-12 vdc min.	_____
	-K voltage		-12 vdc min.	_____
	+K voltage		-12 vdc min.	_____

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		Reference	Limits	Reading
30.2.12	Passive mode tests (Cont'd)	4.5.3.12		
	PITCH DATA voltage UL and DR	4.5.3.12.2	Shall not exceed 100 mv	_____
	YAW DATA voltage UL and DR		Shall not exceed 100 mv	_____
	VALID ECHO voltage		Shall not exceed 200 mv	_____
	-K voltage		Shall not exceed 200 mv	_____
	+K voltage		Shall not exceed 200 mv	_____
	Oscilloscope pulse at J2101-H	4.5.3.12.3	-12 volt min. pulse	_____
	Oscilloscope pulse at J3201-E		-12 volt min. pulse	_____
30.2.13	Fire control test	4.5.3.13		
	TUBE LAUNCH PROGRAM	4.5.3.13.1	-14.7 to -15.3 vdc	_____
	SNAKE SEARCH switch OFF	4.5.3.13.2	-14.7 to -15.3 vdc	_____
	SNAKE SEARCH switch ON			
	SNAKE SEARCH lamp	4.5.3.13.3	Shall illuminate	_____
	SNAKE SEARCH voltage		+0.2 to -0.2 vdc	_____

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		Reference	Limits	Reading
30.2.13	Fire control test (Cont'd)			
	LAUNCH PROGRAM switch to AIR/ASROC	4.5.3.13.4		
	SNAKE SEARCH switch ON		+4.5 to +5.5 vdc	_____
	SNAKE SEARCH switch OFF		+1.0 to -1.0 vdc	_____
	ACTIVE position at J2101-k	4.5.3.13.5	-14.7 to -15.3 vdc	_____
	ACTIVE position at J3201-AA	4.5.3.13.6	+1.0 to -1.0 vdc	_____
	PASSIVE MODE position at J3201-AA	4.5.3.13.7	-14.7 to -15.3 vdc	_____
	PASSIVE MODE position at J2101-k	4.5.3.13.8	+1.0 to -1.0 vdc	_____
	PASSIVE MODE position at J3201-CC	4.5.3.13.9	-14.7 to -15.3 vdc	_____
30.2.14	ODN trigger test	4.5.3.14		
	ODN TRIGGER switch OFF	4.5.3.14.1	-14.7 to -15.3 vdc	_____
	ODN TRIGGER switch ON	4.5.3.14.2	+0.2 to -0.2 vdc	_____
30.2.15	Pitch range constraint test	4.5.3.15		
	PITCH RANGE CONSTRAINT switch ON	4.5.3.15.1	-14.7 to -15.3 vdc	_____
	PITCH RANGE CONSTRAINT switch OFF	4.5.3.15.2	+1.0 to -1.0 vdc	_____

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		Reference	Limits	Reading
30.2.16	Computer enable test	4.5.3.16		
	Negative step to -15v	4.5.3.16.1		_____
	COMPUTER ENABLE lamp	4.5.3.16.2	Shall illuminate	_____
	COMPUTER ENABLE switch OFF		+0.2 to -0.2 vdc	_____
30.2.17	Test jack continuity	4.5.3.17		
	+YAW	4.5.3.17.1	56 K to 62 K	_____
	-YAW		56 K to 62 K	_____
	+PITCH		56K to 62 K	_____
	-PITCH		56 K to 62 K	_____
	ANDED INVERTER		70 K to 80 K	_____
	NOT SAMPLE KNOCKOUT		27 K to 33 K	_____
30.2.18	Test jack continuity			
	R > B WARSHOT to J2101-P	4.5.3.17.2	1 ohm max.	_____
	STEERING READOUT to J2101-d		1 ohm max.	_____
30.2.19	Test jack continuity			
	R < B to MINUS	4.5.3.17.3	2.7 K to 3.3 K	_____
30.2.20	MOD 2 function test			
	+YAW to J2101-X	4.5.3.18.1	56 K to 62 K	_____
	+PITCH to J2101-X		56 K to 62 K	_____
	-PITCH to -15 VDC	4.5.3.18.2	2.15 K to 2.25K	_____
	-YAW to J2101-X	4.5.3.18.3	1 ohm max.	_____
	-YAW to V E INHIBIT	4.5.3.18.4	1 ohm max.	_____
	J3201-R to J3201-CC	4.5.3.18.5	35 K to 37 K	_____
	J2101-C to J3201-AA	4.5.3.18.6	2.64 K to 2.76 K	_____

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		Reference	Limits	Reading
30.2.21	Test jack continuity			
	ANDED INVERTER common jack to the 2 other COMMON jacks	4.5.3.19	1 ohm max.	_____
	J2101-X to J3201-J		Greater than 4 K ohms	_____
30.2.22	Cable continuity			
	+15V to J3201-R	4.5.3.20	1 ohm max.	_____
	-15V to J2101-c		1 ohm max.	_____
	COM to J3201-J		1 ohm max.	_____
	V E INHIBIT to J2101-J		1 ohm max.	_____
	VALID ECHO to J3201-C		1 ohm max.	_____
	PASSIVE MODE to J2101-L		1 ohm max.	_____
	BLANK to J2101-B		1 ohm max	_____
	SAMPLE to J2101-C		1 ohm max	_____
	PING DRIVE to J2101-A		1 ohm max	_____
	RANGE GATE to J2101-E		1 ohm max	_____
	RECEIVER PURSUIT HOLD to J2101-D		1 ohm max	_____
	REINSERT PENDULUM to J3201-X		1 ohm max	_____
	NOT REATTACK to J2101-F		1 ohm max	_____
	R < C to J2101-S		1 ohm max	_____
	R > B to J2101-b		1 ohm max	_____

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		Reference	Limits	Reading
30.2.22	Test jack continuity (Cont'd)			
	PITCH ENABLE to J3201-N		1 ohm max	_____
	COMPUTER ENABLE to J3201-W		1 ohm max	_____
	SNAKE SEARCH to J3201-Z		1 ohm max	_____
	ODN INITIATE to J2101-R		1 ohm max	_____
	SHIFT BEAM AXIS J2101-N		1 ohm max	_____
	BEAM WIDTH FUNCTION to J2101-M		1 ohm max	_____
	NOT SAMPLE KNOCKOUT to J2101-K		1 ohm max	_____
	R < B to J3201-H		1 ohm max	_____
	END SNAKE to J3201-BB		1 ohm max	_____
	AMPLITUDE THRESHOLD to J3201-D		1 ohm max	_____
	YAW DATA to J3201-A		1 ohm max	_____
	PITCH DATA to J3201-B		1 ohm max	_____
	+YAW to J3201-S		1 ohm max	_____
	-YAW to J3201-T		1 ohm max	_____
	+PITCH to J3201-U		1 ohm max	_____
	-PITCH to J3201-V		1 ohm max	_____
	ANDED INVERTER to J3201-L		1 ohm max	_____
30.3	Visual inspections	4.5.4		
30.3.1	Identification marking (major)	4.5.4.1		
	Nomenclature, control and identification callouts	4.5.4.1.1		_____

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		Reference	Limits	Reading
30.3.2	Interchangeability	4.5.4.2		
	Connector receptacles	4.5.4.2.1		_____
30.4	Process verification	4.5.5		
30.4.1	Lacing (minor)	4.5.5.2		
	Lacing and cable configuration	4.5.5.2.1		_____
30.4.2	Soldering (minor)	4.5.5.3		
	Soldered joints	4.5.5.3.1		_____
30.4.3	Cleanliness (major)	4.5.5.4		
	Foreign material	4.5.5.4.1		_____
30.5	Packing and packing inspection	4.5.6		

Preparing Activity
Navy - OS
Project No. 1355-N291

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C-82449A(OS) Computer Test Set (for Torpedo Mk 46)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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Naval Ordnance Station
Indian Head, Md. 20640

OFFICIAL BUSINESS

Commanding Officer
Naval Ordnance Station
Standardization & Data Management Division (QA3)
Indian Head, Maryland 20640

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